Rohit Sinha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
INTERNATIONAL PUBLIC SCHOOL	Class XII (CBSE)	2018	91.40%
D.A.V PUBLIC SCHOOL GANDHI NAGAR	Class X (CBSE)	2016	10.00%

EXPERIENCE

Siemens EDA | Software Engineer Intern

- Refined and improved an existing optimization algorithm within the QuestaSim engine, improving execution performance and simulation efficiency.
- Resolved simulator bugs and developed targeted test cases, performing in-depth debugging to strengthen platform stability and reliability.
- Automated debugging workflows by developing end-to-end Perl automation suites, significantly reducing root-cause analysis turnaround time.
- **Tech Stack:** *C, Perl, GDB, System Verilog, Perforce*

THESIS (ONGOING)

Hardware Security Evaluation of AMD SEV-SNP

- Assessing AMD SEV-SNP confidential computing guarantees across newer AMD EPYC architectures.
- Implemented and reproduced proof-of-concept attacks(e.g., BreakFAST, STALEUS, and RMPocalypse), analyzing hypervisor confused-deputy primitives, memory incoherency, and cache-flush vulnerabilities targeting the PSP and RMP.
- Exploring potential vulnerability vectors in newer AMD EPYC silicon, assessing whether existing microarchitectural mitigations leave residual attack surfaces.

PROJECTS

LockBench - C++ Lock Contention Benchmarking Suite

- Built a benchmarking suite comparing **4 synchronization primitives** -- std::mutex, TAS, TTAS, and ticket locks--across **1-18 worker threads**, multiple contention levels, and heterogeneous P/E-core configurations.
- Automated **160 trials across 8 workload configurations**, producing **1,740 worker-level measurements** for throughput, p50/p95/p99 acquisition latency, and Jain's fairness index with verified CPU affinity.
- Demonstrated that std::mutex delivered up to **2.3× higher throughput than TTAS** under 18-thread contention, while ticket locks maintained **0.988-1.000 fairness** and achieved the lowest p99 latency in **6 of 8 configurations**.
- **Tech Stack:** *C++17, Linux, Multithreading, Atomics, CPU Affinity, Python*

DNSSEC Validation and Key Lifecycle Analysis

- Developed a five-module DNSSEC toolkit for validating RRSIG, DNSKEY, and DS records and verifying the complete chain of trust.
- Built an iterative DNSSEC resolver that resolves domains from root servers while cryptographically validating every delegation.
- Implemented NSEC/NSEC3 denial-of-existence validation, key-rollover analysis, and a Dockerized tampering demonstration that detects unauthorized DNS-record modifications.
- **Tech Stack:** *Python, Cryptography, DNSSEC, BIND9, Docker Compose, Shell Scripting.*

TransactiWar -- Secure Banking & Money Transfer Web Application

- Implemented authentication and database-backed session management using Argon2id password hashing, 256-bit session tokens, CSRF/HMAC protection, secure cookies, token rotation, and session expiration.
- Prevented session and authentication race conditions using database transactions and row-level locking; implemented atomic rate limiting for user search, profile views, and transfer requests.
- Built secure CSV-to-MySQL account provisioning and paginated user search, while strengthening deployment

through TLS 1.3-only Apache configuration and Docker runtime tuning.
- **Tech Stack:** *PHP 8.2, MySQL 8.0, HTML5, CSS3, JavaScript, Apache 2.4, Docker, Docker Compose, PDO, Argon2id, TLS 1.3.*

myShell -- Modular Unix Shell in C

- Built a Unix shell in C11 with a five-state lexer and recursive-descent parser, supporting quoting, variable expansion, multi-stage pipelines, redirections, background execution, and &&, ||, and ; operators.
- Implemented process and job control using fork, execvpe, pipes, dup2, process groups, terminal handoff, and signal handling, including jobs, fg, bg, Ctrl-C, and Ctrl-Z behavior.
- Developed persistent command history, shell/environment variables, field splitting, tilde expansion, and 12 built-in commands; validated functionality.
- **Tech Stack:** *C11, POSIX/Linux system APIs, GCC, GNU Make, Bash test scripting, glibc*

SKILLS

Languages & Tools: C, C++, Perl, Python, Javascript, HTML, CSS, SQL, Git, Perforce, GNU Debugger

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Advanced Computer Architecture & Software Engineering

ACHIEVEMENTS

- Received Knight Badge for being among Top **5%** coders on Leetcode with a peak rating of **1962**.
- Solved over **1,000** data structures and algorithms problems across LeetCode and Codeforces.